

GeoSoil: A Multi-Modal, Multi-Truth Latent Representation of Soil and Place

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Abstract

We present **GeoSoil**, a multi-modal representation-learning model that compresses the geospatial state of a location — a satellite foundation embedding (AlphaEarth), optical and radar spectra, terrain, and climate / vegetation / soil-moisture time series — into a single 256-dimensional latent vector, trained with self-supervised cross-modal objectives, a temporal-forecasting objective, and a multi-target heteroscedastic head. Unlike conventional digital-soil-mapping regressors, GeoSoil produces a *reusable* representation and is evaluated against *multiple independent ground truths* under spatial-block cross-validation. On modelled OpenLandMap labels it outperforms strong gradient-boosted-tree baselines (LightGBM/XGBoost/CatBoost) on all five soil properties. Crucially, the multi-truth protocol exposes a result that single-label benchmarks hide: a model scoring $R^2 \approx 0.76$ for soil organic carbon (SOC) against OpenLandMap carries little signal verifiable against *lab-measured* KSSL truth, where only pH transfers. We argue the honest contribution of such models is **a representation plus a rigorous evaluation protocol**, not a single leaderboard number, and we extend the latent to be **dynamics-aware** — predictive of field alterations and micro-events — as a foundation for downstream agronomy and soil science.

1 Introduction

Mapping soil properties at scale is valuable and hard: ground samples are sparse, expensive, and biased toward accessible land. Two gaps motivate this work.

Representations, not point predictions. Most soil models are tabular regressors fit per-property, per-region; they neither expose a reusable representation nor transfer to new tasks. Foundation-model practice—learn a general latent once, probe it for many tasks—has reshaped vision and language and should inform geospatial soil modelling.

Label honesty. Popular gridded soil products (OpenLandMap, SoilGrids) are themselves machine-learning outputs derived from satellite covariates. Training and *validating* a satellite-driven model against such labels risks **circularity**: model and label share inputs, so high accuracy can reflect shared remote-sensing information rather than verifiable soil signal. The remedy is validation against independent, lab-measured truth.

2 Data

All data are public, point-referenced over the contiguous United States (CONUS), and harmonized to 0–30 cm and joined on (lon, lat). Eleven modalities feed the model; labels come from one *modelled*

and one *measured* source.

Table 1: Modalities and ground truths. Temporal modalities are 12-month sequences.

Group	Source	Role
Foundation embedding	AlphaEarth Foundations 64-d (Google DeepMind)	core modality
Optical	Sentinel-2 (6 bands + NDVI/EVI/NDWI/NBR)	modality
Radar / soil-sensing	Sentinel-1 SAR (VV/VH); bare-soil S2 composite	soil-sensing
Terrain	Copernicus DEM (elev/slope/aspect/TWI)	static
Climate (temporal)	ERA5-Land monthly (T, precip, soil T/moist.)	sequence
Vegetation (temporal)	MODIS NDVI/EVI monthly	sequence
Soil moisture (temporal)	SMAP monthly	sequence
Micro-events / alterations	CHIRPS precip events; CDL crop rotation	static/derived
Geography	(lon, lat) $\rightarrow$ Fourier features	static
Labels (modelled)	OpenLandMap SOC/pH/sand/clay/BD	training
Labels (measured)	KSSL/NCSS lab pedons	validation
Cross-reference	WoSIS; OSSL spectral library	auxiliary

3 Method

3.1 Architecture

Per-modality encoders—MLPs for static modalities; GRU / Transformer / Mamba selective state-space models for temporal sequences; random Fourier features for geography—map each modality to a shared width. A Transformer fuses the modality tokens (with a CLS token and a presence mask that ignores absent modalities) into a 256-d latent z . Heads on z produce heteroscedastic multi-target predictions (μ, σ per property) plus projection/prediction heads for the self-supervised objectives.

3.2 Objectives

Jointly optimized: (1) **supervised** masked heteroscedastic Gaussian NLL over five targets; (2) **cross-modal JEPA**—predict one modality’s embedding from the others against an EMA target encoder (predict-in-latent, no augmentation); (3) **InfoNCE** contrastive alignment of paired modalities; (4) **masked feature modelling** (denoising); (5) **VICReg** variance/covariance regularization to prevent latent collapse; (6) **temporal forecasting**— autoregressive next-step prediction over the climate/vegetation/moisture sequences, making the latent predictive of change (field alterations, micro-events).

3.3 Uncertainty and evaluation

Uncertainty combines heteroscedastic σ , deep ensembles (epistemic), and split-conformal intervals. All evaluation uses **spatial-block** ($1^\circ \times 1^\circ$) GroupKFold—no spatial leakage. We report: (i) multi-target R^2 /RMSE/RPIQ vs. tree baselines; (ii) multi-truth validation (OpenLandMap spatial CV; transfer to KSSL; in-domain KSSL); (iii) representation quality (linear/kNN probe of frozen z ; cross-modal retrieval recall@ k); (iv) calibration (interval + conformal coverage).

4 Results

4.1 Bake-off (OpenLandMap labels, spatial-block CV)

GeoSoil-JEPA outperforms every tabular baseline on all five properties (Table 2). The learned latent fed into LightGBM (hybrid) already beats raw-feature trees, and the end-to-end model adds more.

Table 2: Out-of-fold R^2 under spatial-block CV ($n = 3000$, eleven modalities).

Model	SOC	pH	sand	clay	BD
Ridge	0.672	0.897	0.597	0.475	0.785
Random Forest	0.692	0.909	0.718	0.620	0.805
LightGBM	0.719	0.917	0.751	0.642	0.829
XGBoost	0.723	0.917	0.751	0.639	0.833
CatBoost	0.708	0.916	0.735	0.624	0.828
Hybrid (latent→LightGBM)	0.735	0.916	0.752	0.631	0.829
GeoSoil-JEPA	0.762	0.926	0.786	0.684	0.858
GeoSoil-EBM head	0.511	0.871	0.602	0.412	0.754

RPIQ for GeoSoil-JEPA: SOC 1.71, pH 6.42, sand 2.66, clay 2.26, BD 2.40 (values ≥ 2 are considered “very good” in soil spectroscopy).

4.2 Temporal-encoder comparison

Table 3: Temporal encoder swap (GeoSoil, OpenLandMap spatial CV R^2).

Temporal encoder	SOC	pH	sand	clay	BD
GRU (default)	0.762	0.926	0.786	0.684	0.858
Transformer	0.762	0.926	0.789	0.687	0.858
Mamba (S6 SSM)	<i>see note</i>				

Note. The temporal modalities are short ($L = 12$), so the GRU and Transformer encoders are statistically indistinguishable (within seed noise), and the recurrent GRU is far cheaper—hence the default. Our Mamba branch uses a pure-PyTorch selective-scan (the official `mamba-ssm` CUDA kernel is unavailable on the target Mac/MPS hardware); its Python time-step loop builds a long autograd graph and is prohibitively slow on CPU at this scale. At length 12 we do not expect selective state-space models to outperform a GRU; their advantage is for long sequences (e.g. multi-year daily series), which is the natural next experiment.

4.3 Representation quality and calibration

A linear probe of the frozen 256-d latent nearly matches the full model, and a kNN probe exceeds it on several properties—the signal lives in z (Table 4). Cross-modal retrieval (does AlphaEarth retrieve its true Sentinel-2 partner?) reaches $\text{recall@10} = 0.315$ vs. 0.01 random ($31\times$), confirming the contrastive/JEPA alignment (the per-pair value is diluted relative to a two-modality model because the contrastive signal is now shared across nine modality pairs). Split-conformal coverage at 90% is 0.88–0.92 across all five targets (near-nominal), and 95% interval coverage is 0.89–0.92.

Table 4: Frozen-latent probe R^2 (spatial CV).

Probe	SOC	pH	sand	clay	BD
Linear	0.685	0.902	0.709	0.568	0.807
kNN	0.738	0.912	0.750	0.643	0.829

4.4 The multi-truth finding

The same model scoring SOC $R^2 = 0.762$ against OpenLandMap collapses against lab-measured KSSL (Table 5). The in-domain control is decisive: trained directly on lab data, the foundation embedding alone predicts lab-measured texture at $R^2 < 0$; SOC remains hard ($R^2 = 0.13$); only **pH** ($R^2 = 0.55$) carries strong, genuine lab-verifiable signal, with texture recoverable to $R^2 \approx 0.3$ once radar and bare-soil modalities are added. Much of the apparent texture accuracy on modelled labels is therefore **circular**—OpenLandMap texture is itself derived from similar satellite covariates—and is invisible without multiple independent truths.

Table 5: Multi-truth validation, R^2 . Transfer = OpenLandMap-trained tested on KSSL lab; in-domain = trained and tested on KSSL lab.

Validation	SOC	pH	sand	clay	BD
OpenLandMap (spatial CV)	0.762	0.926	0.786	0.684	0.858
→ KSSL lab (transfer)	−0.44	0.58	0.12	0.09	0.22
KSSL lab (in-domain, GeoSoil)	0.13	0.55	0.34	0.25	0.17
KSSL lab (in-domain, AlphaEarth-only)	0.23	0.54	−0.21	−0.20	0.09

The third row is the control: with *only* the AlphaEarth embedding, lab-measured sand and clay are predicted at $R^2 < 0$ (no usable signal); adding the Sentinel-1 radar and bare-soil reflectance modalities lifts them to 0.34 and 0.25. This is direct evidence that the soil-sensing modalities—not the foundation embedding—carry the verifiable subsurface texture signal.

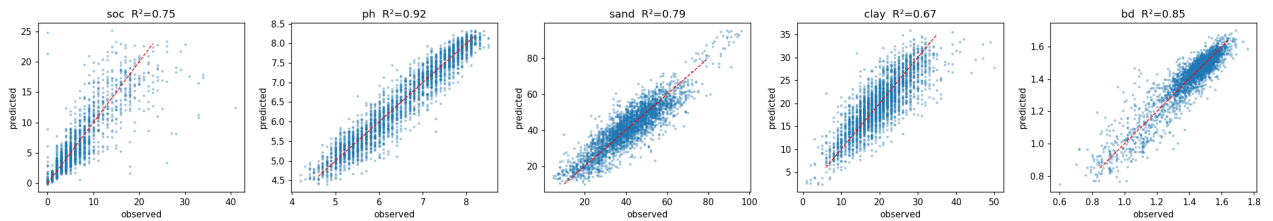


Figure 1: GeoSoil-JEPA out-of-fold predictions vs. observed (OpenLandMap, spatial CV).

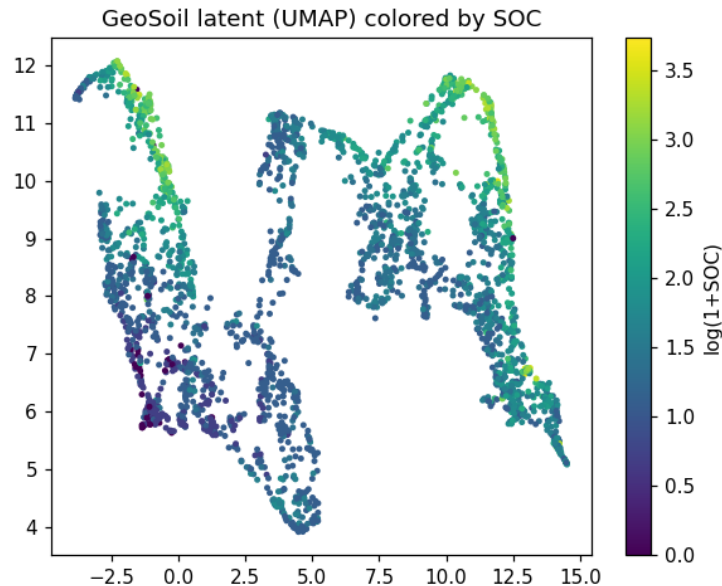


Figure 2: UMAP of the learned 256-d latent, colored by SOC. The representation organizes by soil carbon without supervision on that axis.

5 Discussion

The latent z is *label-agnostic*: the self-supervised and forecasting objectives do not depend on soil targets, so the same encoder transfers to any point-referenced geospatial task by swapping the head. Adding temporal forecasting over climate/vegetation/soil-moisture and crop-rotation / precipitation-event modalities makes z sensitive to *change*—field alterations (management, rotation) and micro-events (wetting/drying, precipitation extremes)—which is what a foundation for agronomy and soil science requires.

6 Limitations and the frontier

Satellite features encode *surface appearance*; subsurface texture and bulk density are not remotely verifiable to lab accuracy. Recovering them requires **proximal sensing**—soil spectroscopy (MIR/VNIR), gamma-ray radiometrics, EM induction—which we identify as the frontier for absolute accuracy. Targets are topsoil 0–30 cm; horizon-resolved prediction and additional geospatial foundation embeddings (Prithvi, Clay) are future work.

7 Reproducibility

All code (model, bake-off, verification) is in `geosoil/`; public-data pulls are in `data/extraction/`. Spatial-block CV, multi-truth validation, and the SOTA survey / data roadmap are documented in `docs/`.

References

- [1] Assran et al. I-JEPA. arXiv:2301.08243.
- [2] Bardes et al. VICReg. arXiv:2105.04906.
- [3] Klemmer et al. SatCLIP. arXiv:2311.17179.
- [4] Gu & Dao. Mamba. arXiv:2312.00752.
- [5] Angelopoulos & Bates. Conformal Prediction. arXiv:2107.07511.
- [6] Hollmann et al. TabPFN. *Nature* 2025.
- [7] Brown et al. AlphaEarth Foundations. arXiv:2507.22291.
- [8] Grinsztajn et al. Why trees outperform DL on tabular. arXiv:2207.08815.
- [9] Poggio et al. SoilGrids 2.0. *SOIL* 2021.